PORA ACADEMY COHORT 7 | PHASE 2C CAPSTONE | GROUP 1

Customer Satisfaction Dashboard: Capstone Report

Olist Brazilian E-Commerce data, 2016 to 2018. Live app: https://pora-customer-satisfaction-dashboard-wu3ymdqefkytau38tt7mss.streamlit.app/

1. Executive summary

We built a Customer Satisfaction dashboard on the Olist Brazilian e-commerce data. It shows how delivery performance links to the review scores customers give. The app has four tabs (Overview, Customer Satisfaction, Detail & Method, Recommendations) and a Year and State filter that changes every KPI and chart.

2. Datasets used

We used the public Olist dataset (8 CSV files). Our notebook loads all 8. The deployed app loads the 5 it needs (orders, customers, order_items, payments, reviews), and our questions mostly use three of them.

Table	Rows	Role in our analysis
orders	99,441	Order status and dates (purchase, delivered, estimated). Basis for KPIs, lateness and delivery days.
order_reviews	99,224	Review score (1 to 5). These rows cover 98,673 distinct orders.
customers	99,441	Gives each order a state. Only 96,096 distinct customer_unique_id values.
order_items, order_payments, products, sellers, category translation	112,650 / 103,886 / 32,951 / 3,095 / 71	Not used in our charts (products, sellers and category translation are loaded in the notebook only).

The geolocation file (about 1,000,000 rows) was left out on purpose. It is very large, slows the deployed app, and none of our charts need it.

3. Methodology

- **Python (pandas):** loads the CSVs and reads query results.
- **SQLite:** in-memory database. Every KPI and chart is a SQL query, with Year and State filters passed as ? placeholders (no f-strings).
- **Streamlit:** the dashboard (filters, tabs, one function per chart, cached connection).
- **Git and GitHub:** team version control. Streamlit Community Cloud deploys from the repo.

How we handled the data.

- No rows were removed or edited. A notebook check function flags quality problems and deletes nothing.
- Reviews repeat for some orders (99,224 rows, 98,673 orders). We kept them all and count orders with COUNT(DISTINCT order_id). Average scores are per review.
- 8 delivered orders have no delivery date. We exclude them from the on-time/late and delivery-time charts, because otherwise SQL counts them as "on time" and "30+ days" (our first notebook gave 88,171 on time; the dashboard gives 88,163).
- The November 2016 gap (no orders) is left as it is. The dashboard does not report GMV; if quoted, we use the all-orders figure, R$15,843,553.24.

Check: with both filters on All, the dashboard shows 99,441 orders, 96,478 delivered, 4.09 average review, 7,826 late and 12.56 average delivery days, matching the reference table.

4. Key findings

Finding 1: Late delivery costs 1.73 stars. On-time orders average 4.294 (88,163 orders, 10.9 days to deliver). Late orders average 2.566 (7,661 orders, 31.4 days). In total 7,826 delivered orders (8.1%) arrived late, on average 9.55 days past the estimate and 189 days at worst. (7,661 of the 7,826 late orders have a review, which is why the two numbers differ.)

Finding 2: Scores fall as delivery takes longer. The biggest single drop (0.53 stars) comes between 11 to 20 days and 21 to 30 days.

0 to 5 days	6 to 10 days	11 to 20 days	21 to 30 days	30+ days
4.45 (n=13,374)	4.36 (n=32,885)	4.22 (n=35,547)	3.69 (n=9,588)	2.26 (n=4,430)

Finding 3: The problem is concentrated by state. RJ has the highest late rate at 13.5% (1,664 of 12,350 delivered orders), about 1.7 times the 8.1% overall rate. SP (5.9%), MG (5.6%), RS (7.1%) and PR (5.0%) are lower. RJ also scores 3.88 across 12,687 reviewed orders. BA is lowest at 3.86 (3,340 orders). PR (4.18) and SP (4.17) are highest.

Other patterns. Reviews are polarised: 57.8% are 5 stars and 11.5% are 1 star. Monthly average score dipped to 3.91 in November 2017 (7,534 reviews, the order peak, likely Black Friday but not proven) and was lower again in February and March 2018 (3.83, 3.75). Months with fewer than 30 reviews (Sep 2016, Dec 2016, Sep and Oct 2018) are too small to read.

5. Recommendations

Recommendation	Evidence and action	Lead
R1 (High): Investigate RJ delivery failures	RJ late rate 13.5% vs 8.1% overall. Audit carriers, warehouses and last-mile routes, then pilot before scaling.	Logistics / Operations
R2 (High): Intervene before day 20	Score drops from 4.22 to 3.69 after day 20. Ageing-order alerts, escalation and revised ETAs.	Logistics + Customer Experience
R3 (Medium): Recalibrate estimated delivery dates	Late orders arrive 9.55 days after the estimate on average, so estimates look too optimistic. Fit ETAs from past transit times by carrier and state.	Data / Analytics + Logistics
R4 (Medium): Pilot improvement in Bahia	BA scores 3.86, the lowest of any state with 2,000+ orders. Reuse the RJ checklist at small scale.	Regional Operations + CX
R5 (Strategic): Promise reliability, not speed	On time 4.29 vs late 2.57: the damage is in the miss. Align delivery messaging with what the network can meet.	Marketing + Logistics

6. Limitations

- The results show association, not causation.
- 2016 and 2018 are partial years, so only 2017 is complete.
- Reviews are voluntary and 768 orders have none.
- No delivery distance data, and results cover Olist in Brazil (2016 to 2018) only.

7. Conclusion

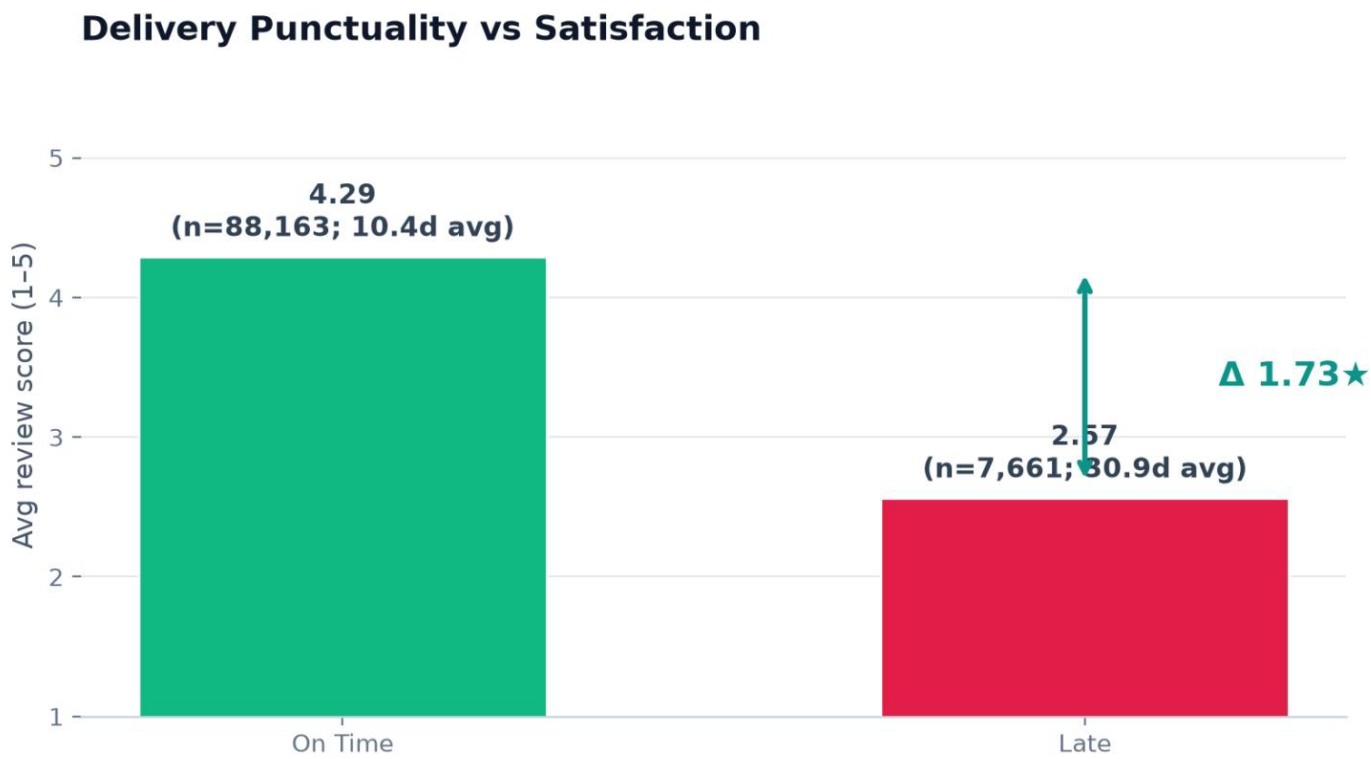
Delivery reliability is the clearest driver of customer satisfaction in this data. On-time orders score 4.29 and late ones 2.57, scores fall steeply once delivery passes 20 days, and RJ stands out with a 13.5% late rate against 8.1% overall. These results come from observed data, so they show where to look first, not proven causes. Our dashboard reproduces the reference figures at All filters and lets users narrow by year and state.

8. Next steps

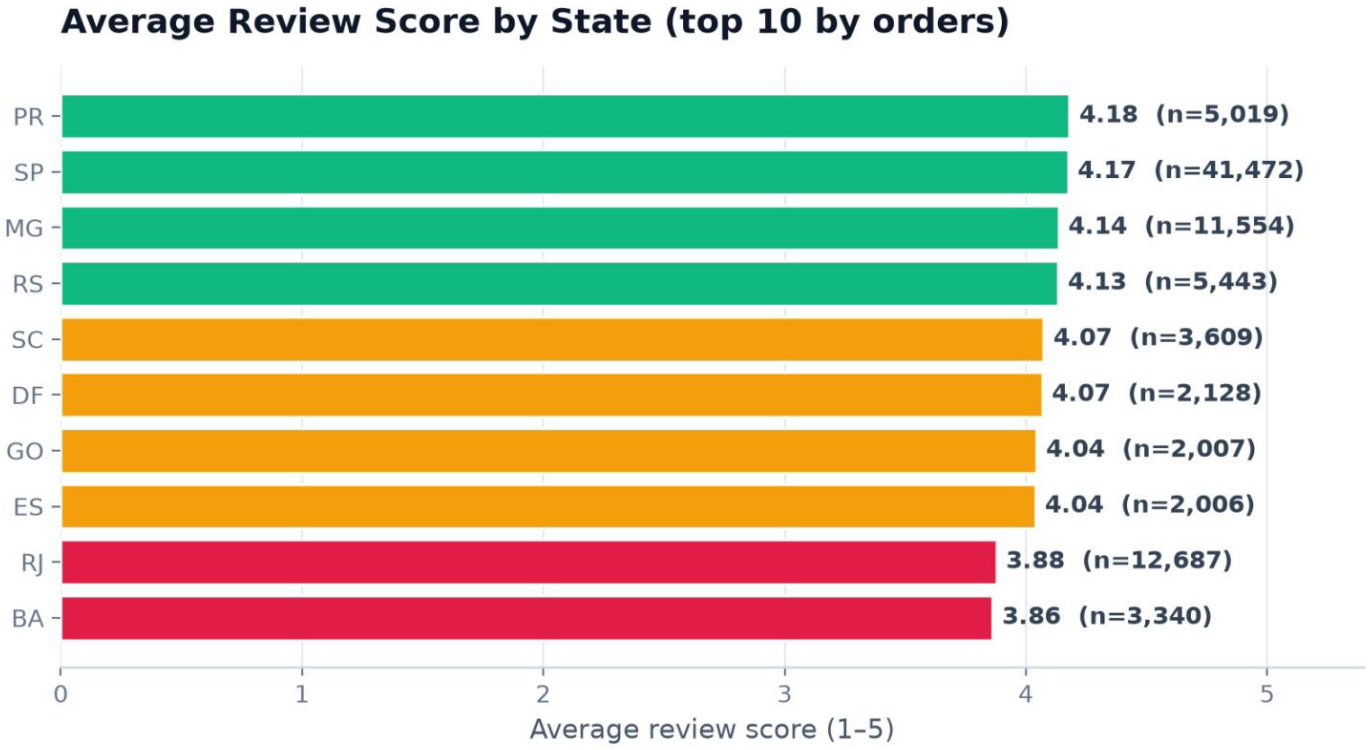
- Run a regression that controls for seller, product category, state and order value to estimate the

lateness effect.

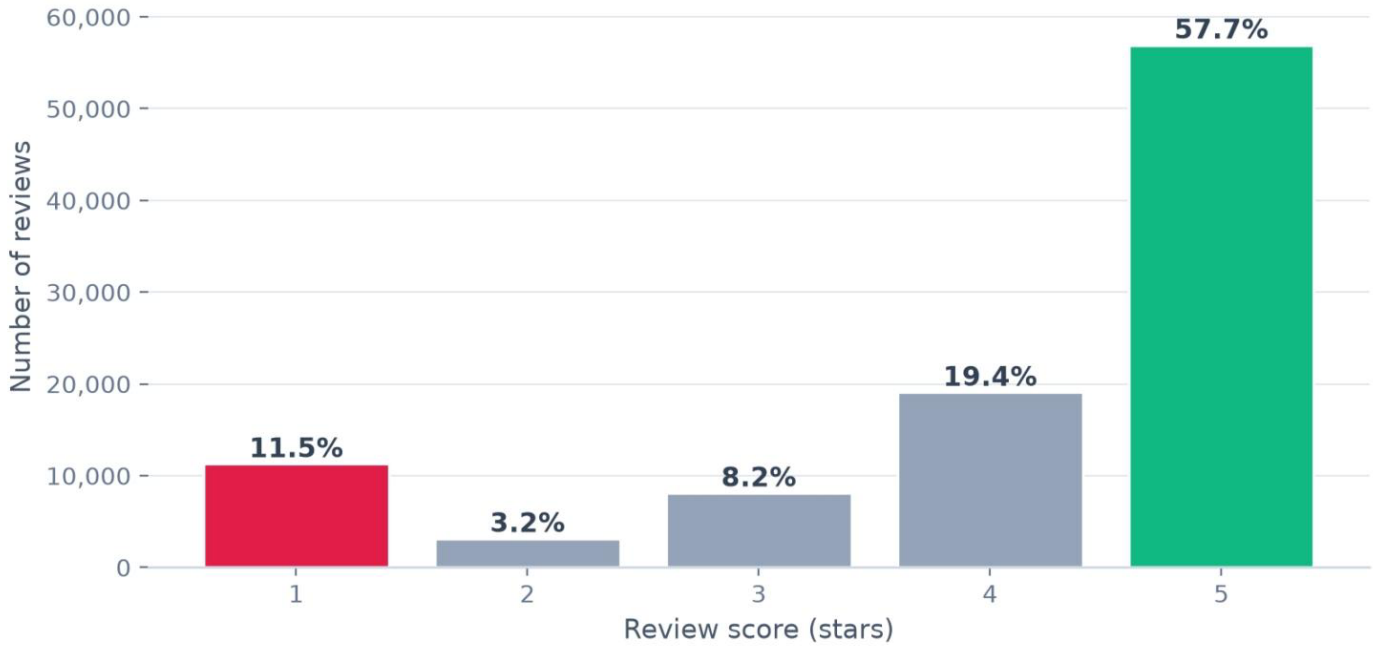
- Split seller delay from carrier delay using the approval, carrier hand-off and delivery timestamps.
- Add delivery distance using the geolocation file and test whether it explains RJ and BA.
- Read and group the low-star review comments (Portuguese) by theme, for RJ and BA first.
- Look into the November 2017 and February to March 2018 dips against order volume and carrier performance.



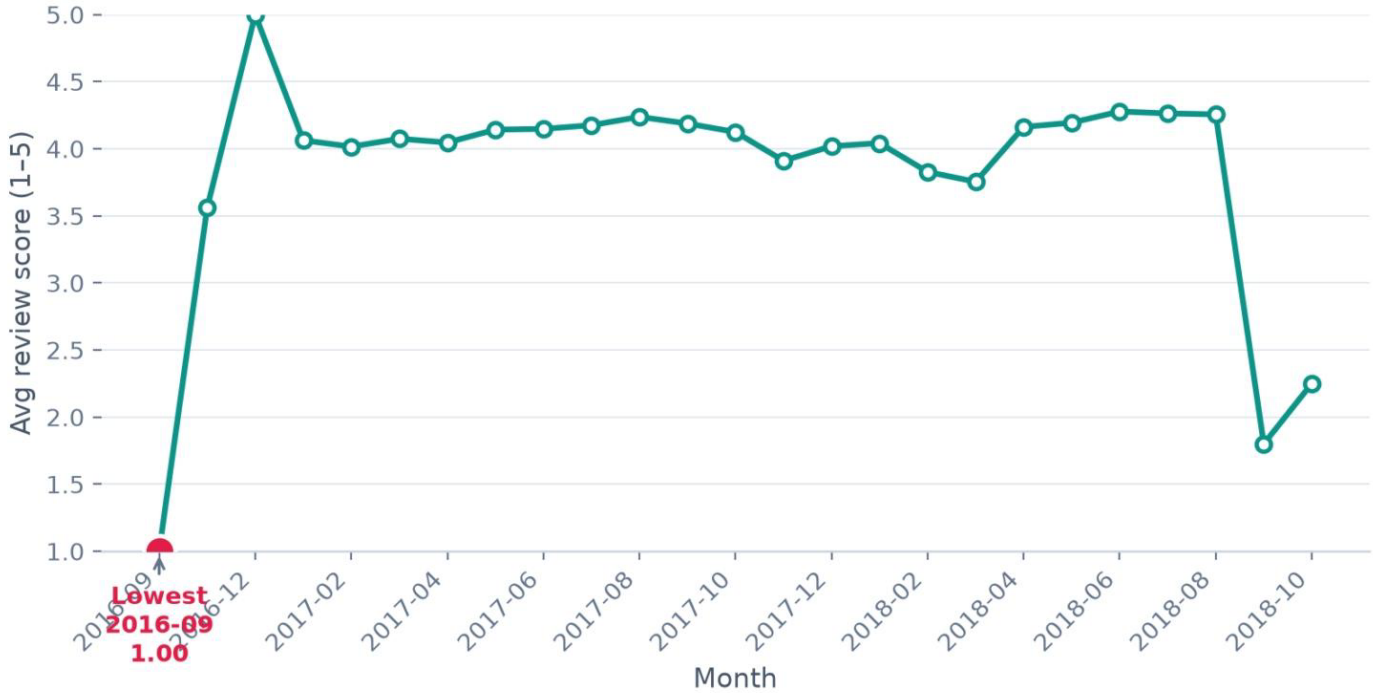
Pilot the RJ fixes and measure against a baseline before claiming any improvement.

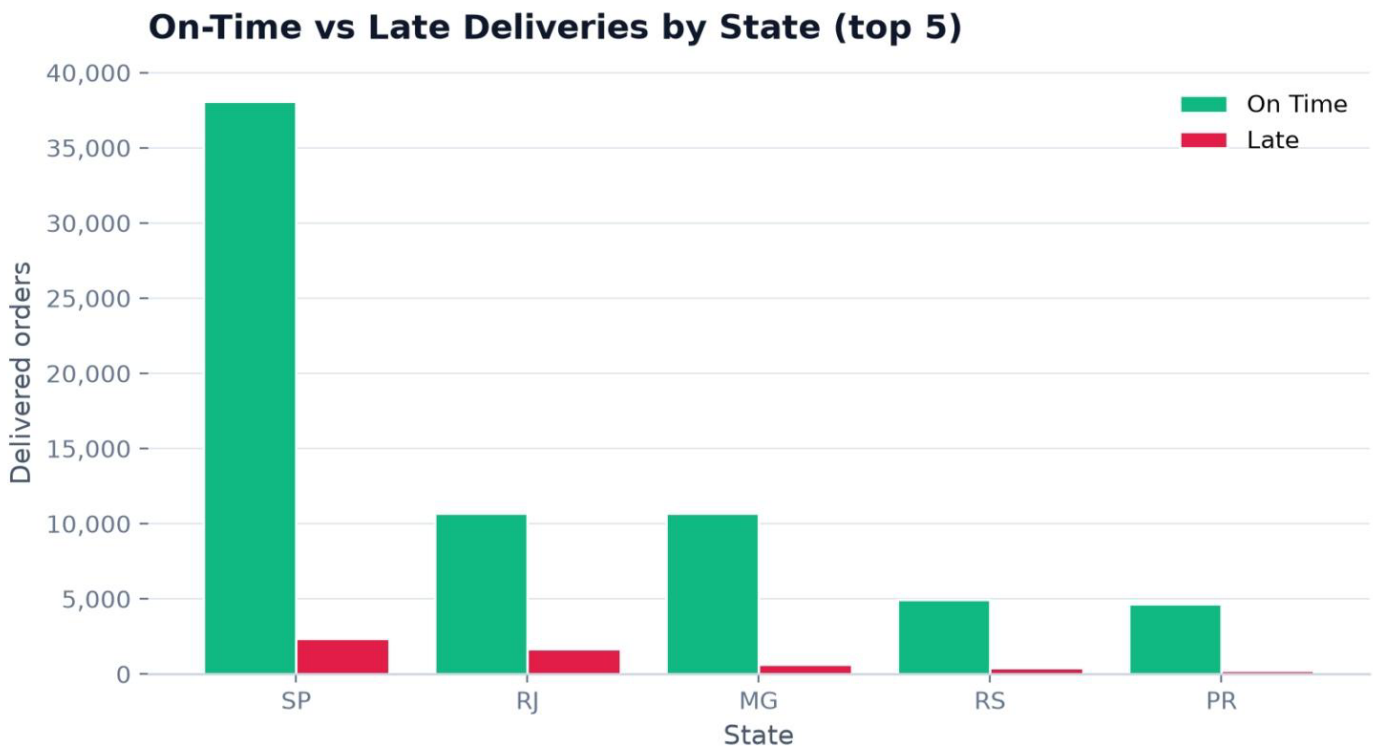


Review Score Distribution



Monthly Average Review Score




On-Time vs Late Deliveries by State (top 5)
40,000
35,000
30,000
25,000
20,000
15,000
10,000
5,000
0
Delivered orders
On Time
Late
SP
RJ
MG
RS
PR
State